SPECTRAL THEORY, SELF-ADJOINT OPERATORS

JOHN E. HUTCHINSON

Contents

1. Definitions

I follow mainly [RSN90].

Let $\mathcal{H}$ be a complex Hilbert space.
Assume $A : \mathcal{D}(A)\ (\subset \mathcal{H}) \to \mathcal{H}$ with A is a linear operator with domain $\mathcal{D}(A)$ dense in $\mathcal{H}$.

(1) A is *closed* if $f_n \to f$, $(f_n)_{n\geq 1} \subset \mathcal{D}(A)$ and $Af_n \to g$, imply $f \in \mathcal{D}(A)$ and $Af = g$.
(2) A is *symmetric* if
$$(Af, g) = (f, AG)\ \forall f, g \in \mathcal{D}(A).$$
It follows that (Af, f) is real for all $f \in \mathcal{D}(A)$.
(3) The *adjoint* A^* of A is defined by
$$\mathcal{D}(A^*) = \{g : \exists h \text{ s.t. } (Af, g) = (f, h)\ \forall f \in \mathcal{D}(A)\},$$
$$A^* g = h \text{ for each such } g \in \mathcal{D}(A^*).$$
(4) If A is symmetric, then the adjoint A^* extends A, written $A \subset A^*$.
A is *self-adjoint* if $A = A^*$.
In particular, A is symmetric, which implies $A \subset A^*$. But the point is that for A to be s.a., the adjoint extension is A itself.

2. Basics of bounded self-adjoint operators

In this section, all operators are *bounded self-adjoint* and in particular everywhere defined.

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2.1. **Bounds.** Let

$$m = \min\{(Af, f) : \|f\| = 1\},$$
$$M = \max\{(Af, f) : \|f\| = 1\}.$$

Then

$$\|A\| = \max\{|m|, |M|\},$$

(easy, [RSN90, p230]),

By $A \leq B$ for two s.a. operators is meant

$$(Af, f) \leq (Bf, f)\ \forall f \in \mathcal{H}.$$

Then

$$m\|f\|^2 \leq (Af, f) \leq M\|f\|^2, \quad \text{i.e. } mI \leq A \leq MI.$$

2.2. **Monotone sequences of s.a. operators.** If A_n are s.a. operators and

$$c_1 I \leq A_1 \leq A_2 \leq \cdots \leq c_2 I$$

then $A_n \to A$ in norm (i.e. strongly) for some bounded s.a. A, [RSN90, p263].

2.3. **Square root of s.a. operators.** If $\mathbb{O} \leq A$ is s.a. then arguing as for (2) below, $\exists$ unique $B =: A^{1/2}$ s.t.

(1) $\mathbb{O} \leq B$ is s.a. and $B^2 = A$.
(2) B is the strong limit of polys in A.
(3) B commutes with all operators that commute with A.

See [RSN90, pp 263–265]

2.4. **Inverse of s.a. operators.** Suppose

$$\mathbb{O} < mI \leq A \leq MI.$$

Then if $M < 2c$, $\|I - cA\| < 1$ and so

$$A^{-1} = c\big(I - (I - cA)\big)^{-1} = cI + cI(I - cA) + c(I - cA)^2 + \cdots,$$

with strong convergence. Moreover,

$$M^{-1}I \leq A \leq m^{-1}I.$$

3. Projections

See [RSN90, pp 266–268].

3.1. **Basics.** If $M \subset \mathcal{H}$ is closed let $P = P_M$ be the (orthogonal) *projection*[1] onto M.
That is, if $f = g + h$ with $g \in M$ and $h \in M^{\perp}$, then $Pf = g$.

It follows

(1) $P^2 = P$, P is s.a., $P_{M^\perp} = I - P$.
(2) Conversely, if $P^2 = P$ and P is s.a. then P is an orthogonal projection. (Note that $\mathcal{R}(P) = \ker(I - P)$ and so is closed)
(3) $(Pf, f) = \|Pf\|^2$.

3.2. **Further properties.** Suppose $P_1 = P_{M_1}$ and $P_2 = P_{M_2}$. Then

(1) P_1 and P_2 commute implies $P_1P_2 = P_{M_1 \cap M_2}$.
(2) $P_1P_2 = 0$ implies

$$P_2P_1 = 0, \quad M_1 \cap M_2 = \{0\}, \quad M_1 \perp M_2, \quad P_1 + P_2 = P_{M_1 \oplus M_2}.$$

(3) P_1 and P_2 commute implies $P_1 + P_2 - P - 1P_2 = P_{M_1 + M_2}$.
(4) $P_1P_2 + P_2$ iff $P_1 \geq P_2$ iff $M_2 \subset M_1$. In this case, $P_1 - P_2 = P_{M_1 \cap M_2^{\perp}}$.

[1] All projections are orthogonal unless otherwise stated.

4. **Functions of a bounded s.a. operator**

4.1. **Polynomials.** Suppose A is bounded s.a. and $mI \le A \le MI$.

If $p = p(\lambda)$ is a polynomial then $p(A)$ *is a well defined s.a. operator* and the correspondence $p \to p(A)$ is

(1) *linear*;
(2) *multiplicative*;
(3) *monotonic*, and in particular *preserves positivity*:

$$p(\lambda) \le q(\lambda) \text{ for } \lambda \in [m, M] \Longrightarrow p(A) \le q(A).$$

Proof: Sufficient to show

$$p(\lambda) \ge 0 \text{ if } \lambda \in [m, M] \Longrightarrow p(A) \ge 0.$$

But under this hypothesis,

$$p(\lambda) = \prod_i (\lambda - a_i) \prod_j (b_j - \lambda) \prod_k \left((\lambda - c_k)^2 + d_k^2\right)$$

where $p(\lambda) = 0$ for $\lambda = a_i < m$, $\lambda = b_j > M$, $\lambda = c_k$ is a double root for $m \le c_k \le M$ and then $d_k = 0$, or $\lambda = c_k \pm d_k$ and then $d_k \ne 0$.

Each of the factors is positive s.a. if λ is replaced by A, and hence so is the product $p(A)$.

(4) $|p(\lambda) - q(\lambda)| \le \epsilon$ on $[m, M]$ implies $\|p(A) - q(A)\| \le \epsilon$.

4.2. **Larger classes of functions.**

(1) Suppose u is u.s.c., i.e. $\limsup_{\mu \to \lambda} f(\mu) \le f(\lambda)$. For example, $u = \mathcal{X}_I$ if I is a *closed* interval. Then $\exists$ polynomials $0 \le p_n \downarrow u$ in $[m, M]$.

Hence $c \le p_n(A) \downarrow$ and so the limit exists by Section 2.2 and is denoted by $u(A)$.

(2) Moreover $cI \le u(A)$ *is s.a.* and independent of the sequence p_n. The *analogous properties to those in Section 4.1* are preserved.
(3) *If $u = v$ on $[m, M]$ then $u(A) = v(A)$.* (From (4) in Section 4.1).)
(4) Similarly if u is the difference of two u.s.c. functions then $u(A)$ is well defined with analogous properties. An example is $u = \mathcal{X}_I$ if I is *any* interval.

5. **Spectral decomposition of bounded s.a. operators**

Assume A is a bounded s.a. operator with $mI \le A \le MI$.

5.1. **Associated projection maps.** For $\lambda \in \mathbb{R}$ define

$$e_\lambda := \mathcal{X}_{(-\infty,\lambda]}, \quad E_\lambda := e_\lambda(A).$$

Then E_λ is (bounded) s.a. operator Moreover, E_λ *is an orthogonal projection onto* M_λ, say.

Proof: $e_\lambda^2 = e_\lambda$ and so $E_\lambda^2 = E_\lambda$.

We often write $E_{(-\infty,\lambda]} = E_\lambda$. More generally, for any interval I,

$$e_I := \mathcal{X}_I, \quad E_I := e_I(A).$$

Then again $E_I : \mathcal{H} \to \mathcal{H}$ with range M_I (say) is a projection, since $e_I^2 = e_I^2$ and so $E_I^2 = E_I$.

Note that $E_{(a,b]} = E_b - E_a$ since $e_{(a,b]} = e_b - e_a$.

In this way E is an operator valued measure defined on Borel subsets of $\mathbb{R}$ and $E_B = E(B)$ are orthogonal projections.

5.2. **Approximate eigenspace and eigenvalue.** Suppose I is an interval with endpoints $\mu \pm \epsilon$. Then

$$(\mu - \epsilon) e_I(\lambda) \le \lambda e_I(\lambda) \le (\mu + \epsilon) e_I(\lambda) \ \forall \lambda$$
$$\therefore (\mu - \epsilon) E_I \le A E_I \le (\mu + \epsilon) E_I$$

Hence $\|AE_I - \mu E_I\| \le \epsilon$ from Section 2.1, and so

$$\|A\phi - \mu\phi\| \le \epsilon \|\phi\| \quad \forall \phi \in \mathcal{M}_I. \tag{1}$$

For small ϵ we can thus think of $\mathcal{M}_I$ as an *approximate "eigenspace" for A with "approximate" eigenvalue* μ.

Let $m = \xi_0 < \xi_1 < \xi_2 < \xi_3 < \cdots < \xi_N = M$ be a partition of $[m, M]$. Let $I_1 = [\xi_0, \xi_1]$ and $I_n = (\xi_{n-1}, \xi_n]$ for $1 < n \le N$.

Then for $\lambda \in [m, M]$

$$1 = \sum_{n=1}^{N} e_{I_n}(\lambda), \quad \lambda = \sum_{n=1}^{N} \lambda e_{I_n}(\lambda), \text{ and so} \tag{2}$$

$$I = \sum_{n=1}^{N} E_{I_n}, \quad A = \sum_{n=1}^{N} A E_{I_n}, \tag{3}$$

Then for $f \in \mathcal{H}$,

$$f = \sum_{n=1}^{N} E_{I_n} f = \sum_{n=1}^{N} (f, \phi_n)\phi_n, \tag{4}$$

$$Af = \sum_{n=1}^{N} A E_{I_n} f \approx \sum_{n=1}^{N} \xi_n (f, \phi_n)\phi_n, \tag{5}$$

where $\phi_n := E_{I_n} f / \|E_{I_n} f\|$, and in case $E_{I_n} f = 0$ then we *drop* the corresponding term in the sum. Note that the unit vector ϕ_n must depend on f unless M_{I_n} is one dimensional.

This gives an "approximate" spectral decomposition of $\mathcal{H}$ into orthogonal "approximate" eigenspaces $\mathcal{M}_n = \mathcal{R}(E_{I_n})$ for A, "approximate" unit eigenfunctions ϕ_n by the previous discussion, and "approximate" eigenvalues ξ_n. Note that ϕ_n depends on f unless $\mathcal{M}_n$ is one dimensional.

5.3. **Spectral Resolution.** Suppose A and notation as in the previous section. If $|I_n| \le \epsilon$ it follows from (1) that

$$\|A - \sum_{n=1}^{N} \lambda_n E_{I_n}\| \le \epsilon, \tag{6}$$

for any $\lambda_n \in I_n$.

From this and the first equation in (3) one can pass to a Riemann Stieltjes limit to get

$$I = \int_m^M dE_\lambda, \quad f = \int_m^M dE_\lambda f \ \forall f \in \mathcal{H}, \quad (f, g) = \int_m^M d(E_\lambda f, g) \ \forall f, g \in \mathcal{H},$$

$$A = \int_m^M \lambda \, dE_\lambda, \quad Af = \int_m^M \lambda \, dE_\lambda f \ \forall f \in \mathcal{H}, \quad (Af, g) = \int_m^M \lambda \, d(E_\lambda f, g) \ \forall f, g \in \mathcal{H},$$

Here E can be considered as either an increasing, operator valued, distribution function $\lambda \mapsto E_\lambda = E_{(-\infty,\lambda]}$ supported on $[m, \infty)$; or as an operator valued measure supported on $[m, M]$. Likewise $E_\lambda f$ is function valued distribution function or measure, and $(E_\lambda f, g)$ is a real distribution function or measure.

Replace λ by λ^r and A by A^r in (6). This is valid for $r = 0$ by (4) and for integers $r \ge 1$ from (6) because of the orthogonality of the E_{I_n}. It follows that

$$u(A) = \int_m^M u(\lambda) \, dE_\lambda$$

for polynomials u, and hence for continuous functions u by an approximation argument.

It follows tthat

$$(u(A)f, g) = \int_m^M u(\lambda) \, d(E_\lambda f, g)$$

because a similar fact is true for the approximating Riemann sums. This gives *uniqueness* of the function $\lambda \mapsto (E_\lambda f, g)$ and hence of E_λ, at points of continuity and hence everywhere by right continuity and density of continuity points.

5.4. **Examples.** For a *compact symmetric s.a. operator* A, setting $L_\phi f := (f,\phi)$, one has

$$A = \sum_{n\geq 1} \lambda_n \phi_n L_{\phi_n}, \quad Af = \sum_{n\geq 1} \lambda_n (f,\phi_n)\phi_n, \quad (Af,g) = \sum_{n\geq 1} \lambda_n (f,\phi_n)\overline{(g,\phi_n)},$$

where $(\phi_n)_{n\geq 1}$ is an orthonormal basis of $\mathcal{H}$ and the e-values $\lambda_n \to 0$.

Writing E as a measure or a distribution function respectively, this corresponds to

$$E = \sum_n \delta_{\lambda_n}\phi_n L_{\phi_n}, \quad E_\lambda = \sum_{\lambda_n\leq\lambda} \phi_n L_{\phi_n},$$

$$Ef = \sum_n \delta_{\lambda_n}\phi_n (f,\phi_n), \quad E_\lambda f = \sum_{\lambda_n\leq\lambda} \phi_n (f,\phi_n),$$

$$(Ef,g) = \sum_n \delta_{\lambda_n}(f,\phi_n)\overline{(g,\phi_n)}, \quad E_\lambda f = \sum_{\lambda_n\leq\lambda} (f,\phi_n)\overline{(g,\phi_n)}.$$

For an example with continuous spectrum consider the multiplier case

$$\mathcal{H} = L^2(0,1), \quad (Af)(x) = xf(x), \quad \text{i.e. } Af = mf,$$

where $m(x) = x$.

It follows $(A^r f)(x) = x^r f(x)$ for all integers $r \geq 0$ and so

$$\big(u(A)f\big)(x) = u(x)f(x),$$

for all polynomials u, and hence for all continuous and semi continuous u.

In particular, $(E_\lambda f)(x) = e_\lambda(x)f(x)$ where $e_\lambda(x) = \mathcal{X}_{(-\infty,\lambda]}$. Hence for *fixed* x, $(E_\lambda f)(x)$ is the function of λ which *jumps* from 0 to $f(x)$ at $\lambda = x$. Hence the measure $dE_\lambda f(x)$ is the *point measure* $f(x)\delta_x(\lambda)$.

Hence $Af = \int_0^1 \lambda\, dE_\lambda f$ becomes

$$xf(x) = \int_0^1 \lambda\, d\big(f(x)\delta_x(\lambda)\big),$$

and the right side also directly gives $xf(x)$.

References

[RSN90] Frigyes Riesz and Béla Sz.-Nagy, *Functional analysis*, Dover Publications Inc., 1990. Translated from the second French edition by Leo F. Boron; Reprint of the 1955 original.